

II Year B.Tech. CSE (AI) – I Semester

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(R23PC31) ARTIFICIAL INTELLIGENCE**Pre-requisite:**

- Knowledge in Computer Programming.
- A course on “Mathematical Foundations of Computer Science”.
- Background in linear algebra, data structures and algorithms, and probability.

Course Objectives:

- The student should be made to study the concepts of Artificial Intelligence.
- The student should be made to learn the methods of solving problems using Artificial Intelligence.
- The student should be made to introduce the concepts of Expert Systems.
- To understand the applications of AI, namely game playing, theorem proving, and machine learning.
- To learn different knowledge representation techniques

Course Outcomes:

- Explain the concepts of AI and Intelligent agents.
- Apply the various search strategies in problem solving.
- Examine the effectiveness of truths by knowledge representation methods in AI.
- Analyze the logical agents to do inference using first order logic.
- Understand the basics of an expert system.

UNIT – I

Introduction: AI problems, foundation of AI and history of AI intelligent agents: Agents and Environments, the concept of rationality, the nature of environments, structure of agents, problem solving agents, problem formulation.

UNIT - II

Searching- Searching for solutions, uniformed search strategies – Breadth first search, depth first Search. Search with partial information (Heuristic search) Hill climbing, A* ,AO* Algorithms, Problem reduction, Game Playing-Adversial search, Games, mini-max algorithm.

UNIT - III

Representation of Knowledge: Knowledge representation issues, predicate logic- logic programming, semantic nets- frames and inheritance, constraint propagation, representing knowledge using rules, rules-based deduction systems, **forward chaining, Backward chaining, Resolution**, Reasoning under uncertainty, review of probability, Bayes’ probabilistic interferences and dempster-shafer theory.

UNIT - IV

Functional Learning: Introduction to Machine Learning, Supervised and Unsupervised learning, Decision trees, Explanation based learning, Statistical Learning methods, Reinforcement Learning, **Applications of AI in healthcare, education and finance.**

UNIT - V

Expert Systems: Architecture of expert systems, Knowledge Acquisition, Typical expert systems – MYCIN, DART, XCON, Hardware based AI-GPU, TPU, NPU, Ethical issues in AI **Introduction to Generative AI and LLM’s, Prompt Engineering, Retrieval-Augmented Generation (RAG).**

Textbooks:

1. S. Russel and P. Norvig, “Artificial Intelligence – A Modern Approach”, Second Edition, Pearson Education.
2. Kevin Night and Elaine Rich, Nair B., “Artificial Intelligence (SIE)”, Mc Graw Hill

Reference Books:

1. David Poole, Alan Mackworth, Randy Goebel,”Computational Intelligence: a logical approach”, Oxford University Press.
2. G. Luger, “Artificial Intelligence: Structures and Strategies for complex problemsolving”, Fourth Edition, Pearson Education.
3. J. Nilsson, “Artificial Intelligence: A new Synthesis”, Elsevier Publishers.
4. Artificial Intelligence, SarojKaushik, CENGAGE Learning.

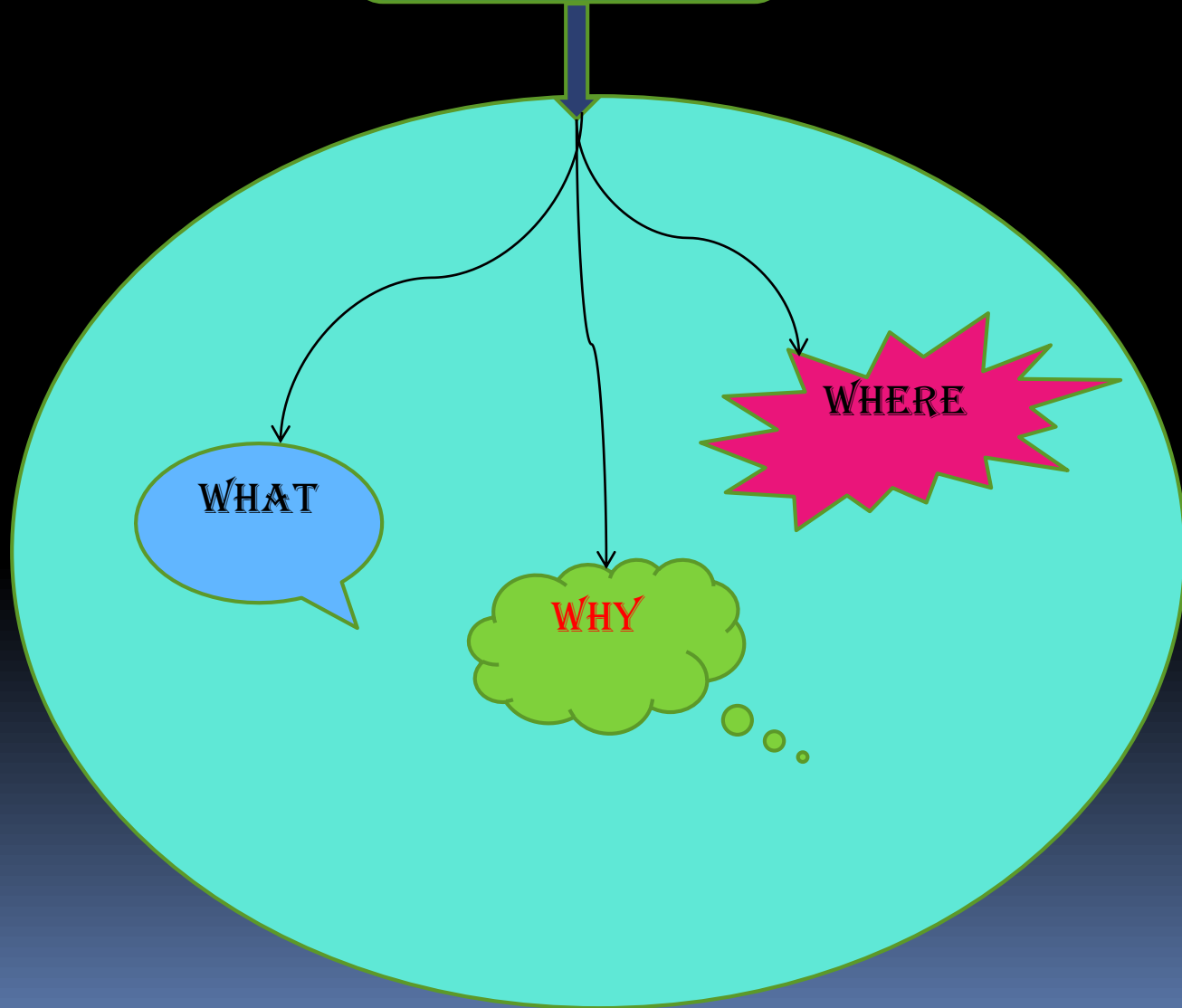
Online Learning Resources:

1. <https://ai.google/>
2. https://swayam.gov.in/nd1_noc19_me71/preview
3. https://onlinecourses.nptel.ac.in/noc22_cs56/preview



INTRODUCTION TO AI

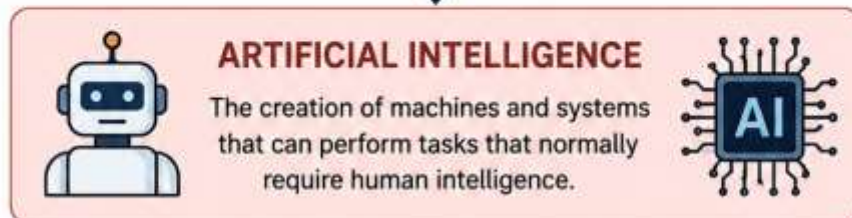
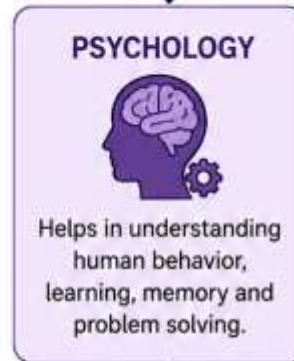
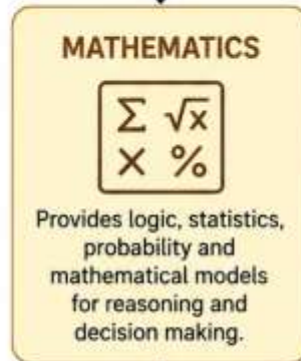
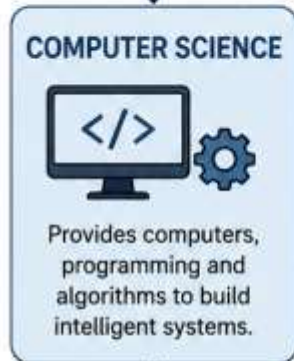
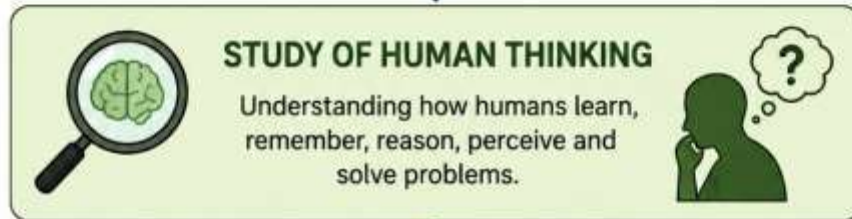
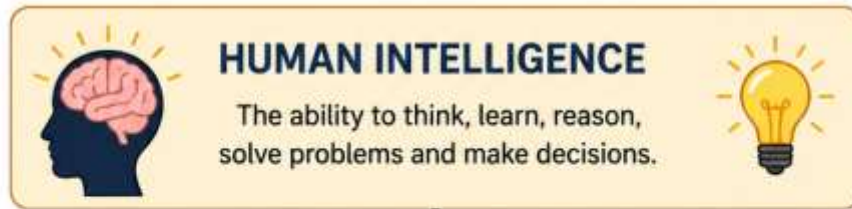
Artificial Intelligence



WHAT

- ❑ **Breaking Down the Term "Artificial Intelligence"**
- ❑ **Artificial** = Made by humans (not natural)
- ❑ **Intelligence** = Ability to learn, think, reason, and make decisions
- ❑ **Therefore,**
- ❑ **Artificial Intelligence** = Human-made intelligence.

➤ **Artificial Intelligence (AI)** is the ability of a computer or machine to **perform tasks that normally require human intelligence.**





Why Was AI Created?

AI was created to make computers capable of:

- Learning from data
 - Solving problems
 - Making decisions
 - Understanding human language
 - Recognizing images and speech
 - Performing tasks that normally require human intelligence
- 

WHERE DID AI COME FROM?

- Artificial Intelligence did not appear suddenly. It developed from several fields of study:
- **Computer Science** – Provides the computers and programming needed to build AI systems.
- **Mathematics** – Supplies algorithms, probability, statistics, and logic for AI.
- **Psychology** – Helps researchers understand how humans learn, think, and make decisions.
- **Neuroscience** – Inspires AI by studying how the human brain works.
- **Linguistics** – Helps AI understand and generate human language.
- **Philosophy** – Explores reasoning, knowledge, and intelligent behavior.
- These fields together led to the development of AI.



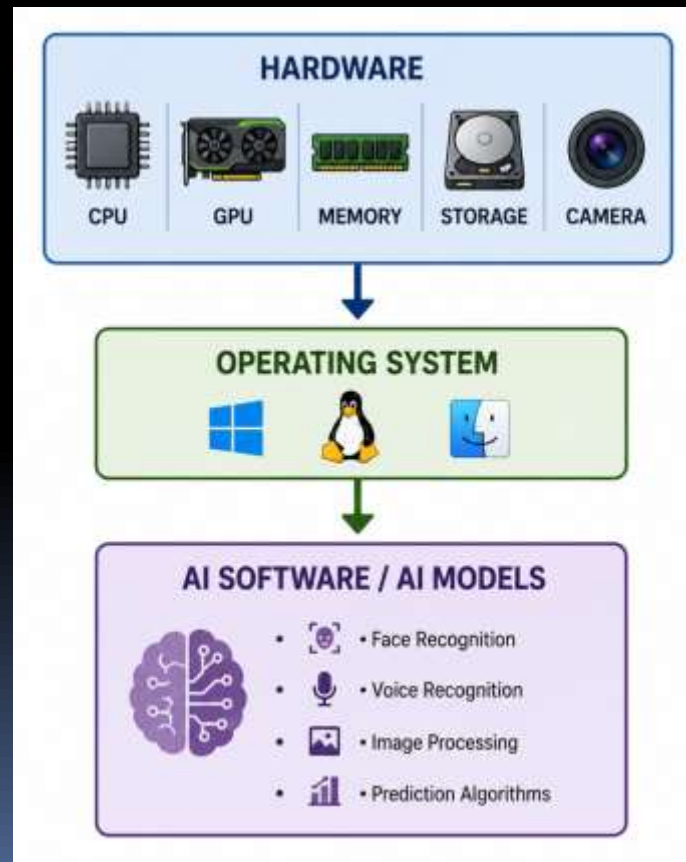
WHO INTRODUCED AI?

John McCarthy introduced the term "Artificial Intelligence" in 1956 during the Dartmouth Conference, which is considered the birth of AI as a formal field of research.



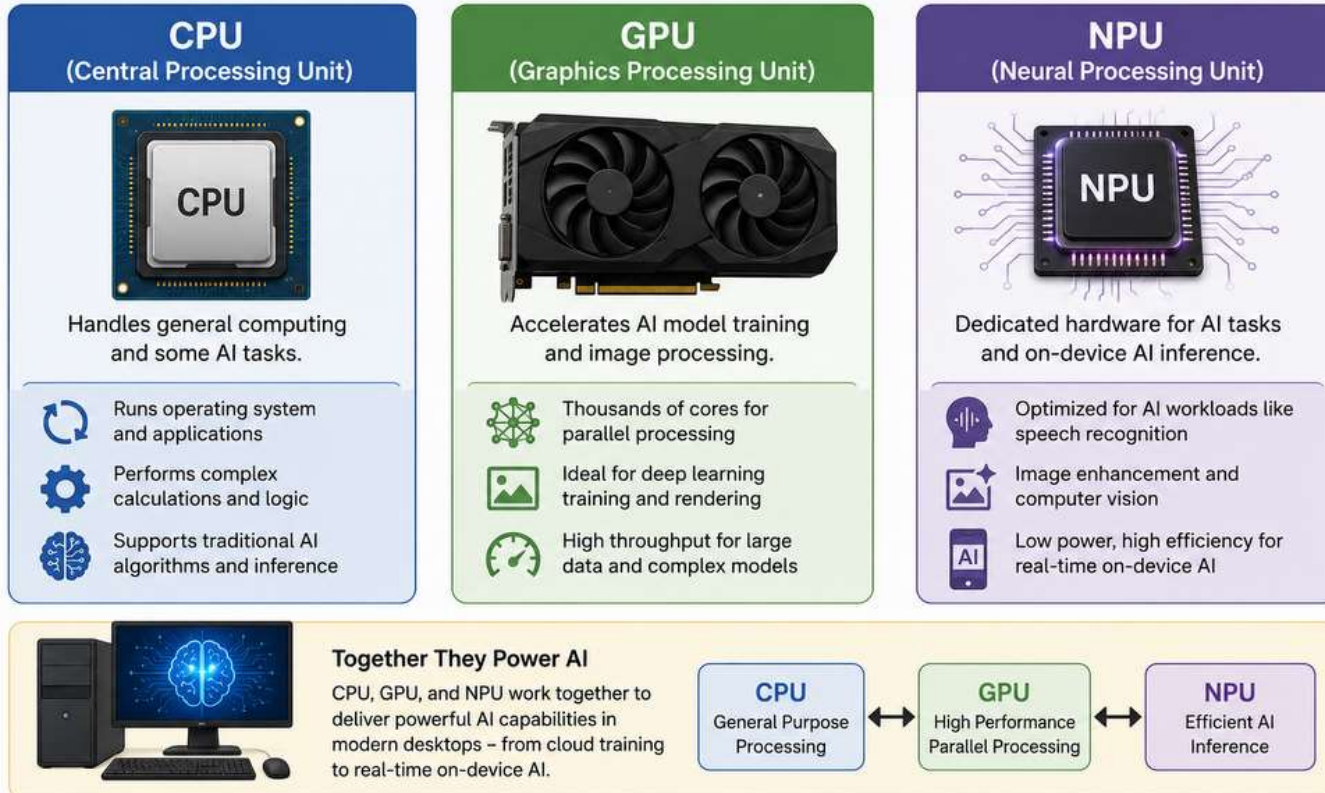
WHERE IS AI PRESENT?

➤ AI is **not** a physical object like a CPU or RAM. It is **software** (algorithms and models) that runs on computer hardware.



AI HARDWARE INSIDE MODERN DESKTOPS

Modern computers include specialized hardware to run AI workloads efficiently.



Everyday Example

Suppose you unlock your phone using your face.

- The **camera** captures your face.
- The **AI model** compares your face with the stored data.
- The **processor (CPU/GPU/NPU)** performs the calculations.
- If it matches, the phone unlocks.

Here:

- Camera = Sensor
- CPU/GPU/NPU = Processing hardware
- AI model = Intelligence

Key Point

- AI is **not inside one specific chip or component**. It is a combination of:
- **Algorithms** (rules and mathematical models)
- **Data** (used for learning)
- **Software** (implements the AI)
- **Hardware** (CPU, GPU, or NPU that executes it)

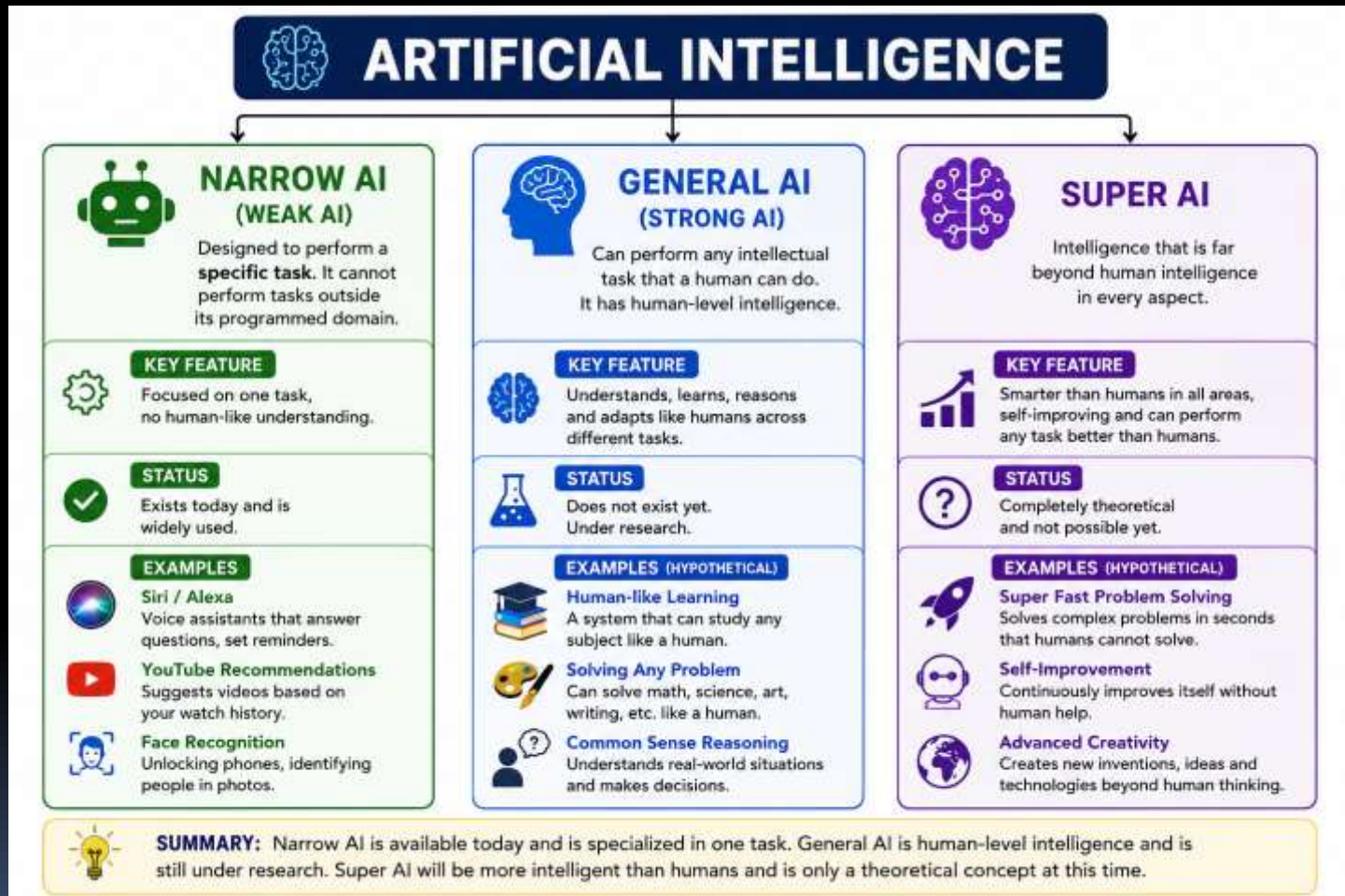


TYPES OF AI

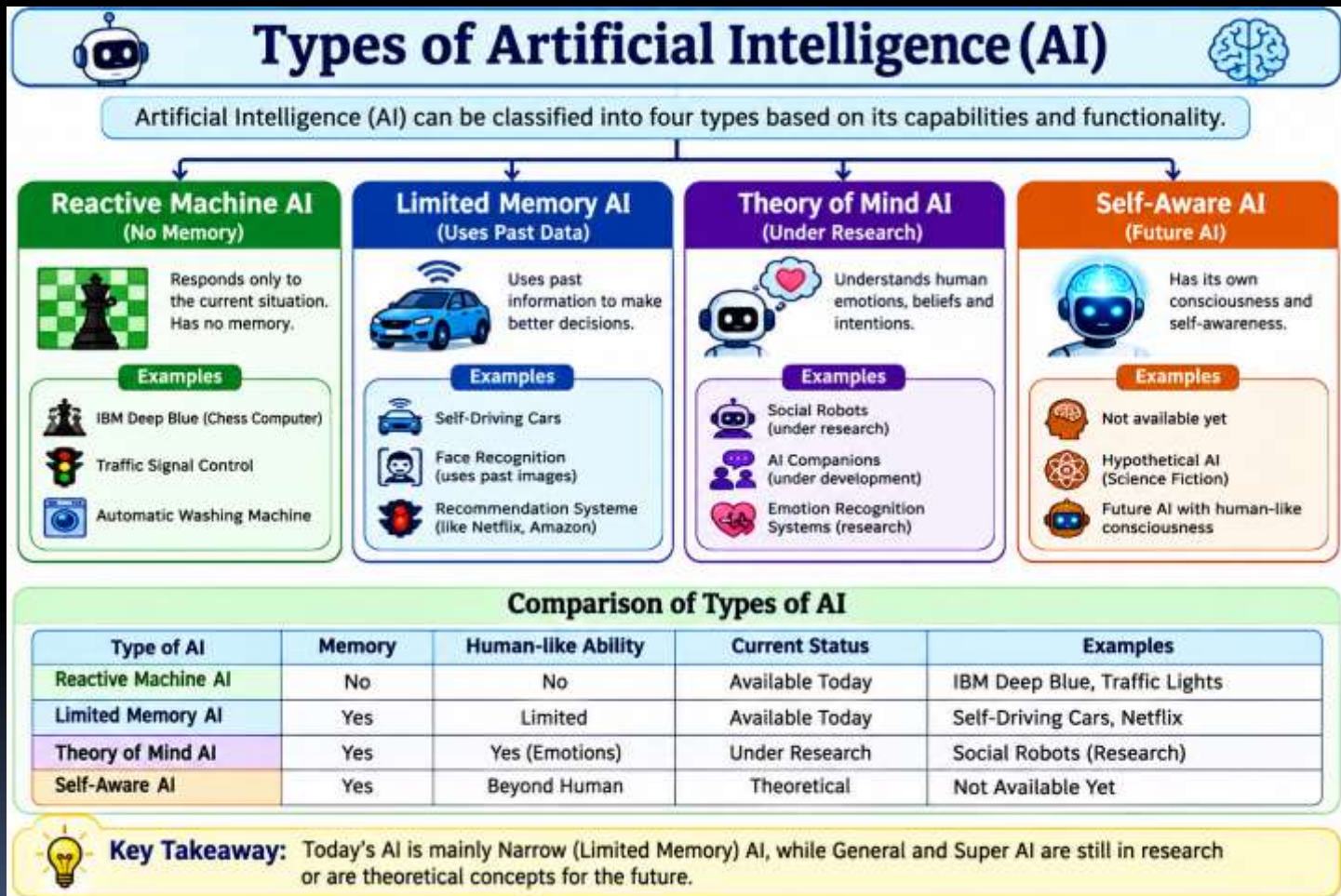
Artificial Intelligence can be classified in **two ways**:

1. **Based on Capabilities**
 2. **Based on Functionality**
- 

BASED ON CAPABILITIES



BASED ON FUNCTIONALITY





AI PROBLEMS

AI faces several challenges and problems, both technical and ethical. Here are some key issues:

Here are the key issues related to AI:

- **Bias and Fairness**
- **Explainability and Transparency**
- **Data Privacy and Security**
- **Ethical Concerns**
- **Job Displacement**
- **Generalization and Robustness**
- **Scalability and Computational Resources**
- **Legal and Regulatory Issues**
- **Human-AI Interaction**
- **Sustainability**

AI PROBLEM

SIMPLE MEANING

EXAMPLE

Bias and Fairness

AI treats some people unfairly

Biased hiring system

Explainability and Transparency

AI cannot explain its decisions

Loan rejection

Data Privacy and Security

Personal data must be protected

Hospital records

Ethical Concerns

AI should make morally correct decisions

Self-driving car

Job Displacement

AI replaces some human jobs

Factory robots

Generalization and Robustness

AI should work in new situations

Self-driving car in rain

Scalability and Computational Resources

AI needs powerful computers

Training large AI models

Legal and Regulatory Issues

AI needs laws and regulations

Responsibility for AI mistakes

Human-AI Interaction

AI should communicate well with people

Customer service chatbot

Sustainability

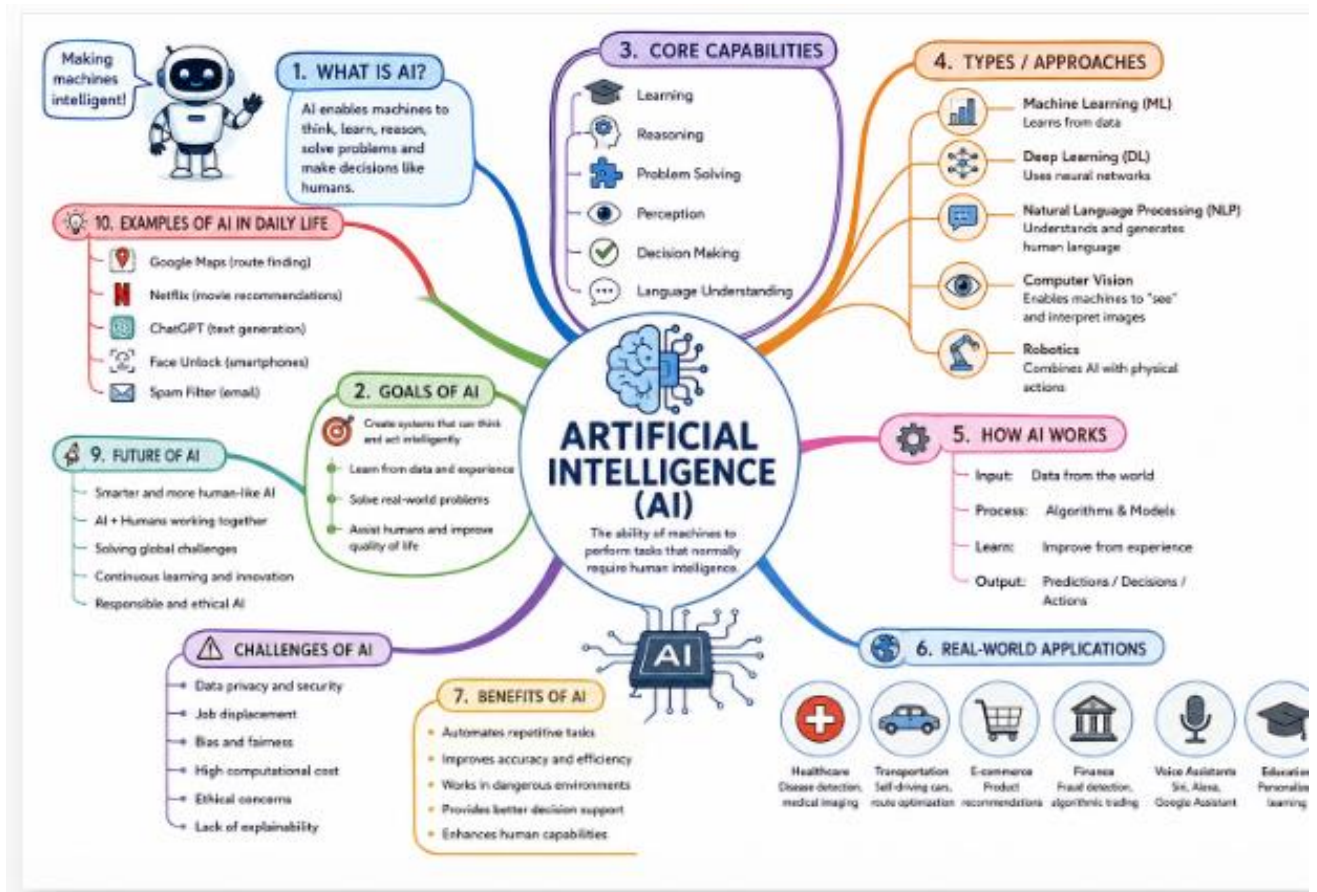
AI should use less energy

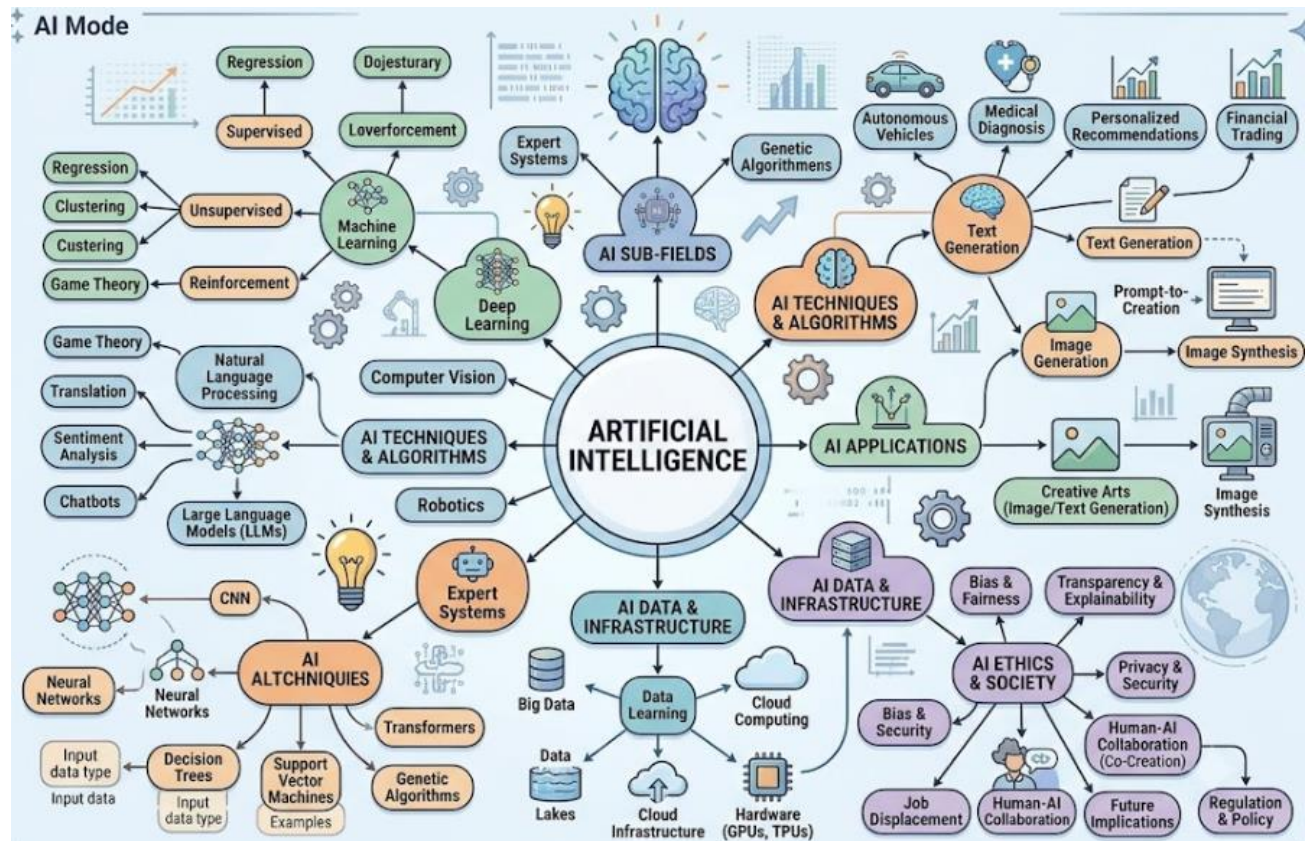
Green AI and efficient data centers

Artificial Intelligence

Inquiry

1. Can a machine think like a human?
2. How does a computer recognize a face or voice?
3. How does Google Maps find the best route?
4. Can a machine learn from its mistakes?
5. Where do we use Artificial Intelligence in daily life?





What is Artificial Intelligence (AI)?

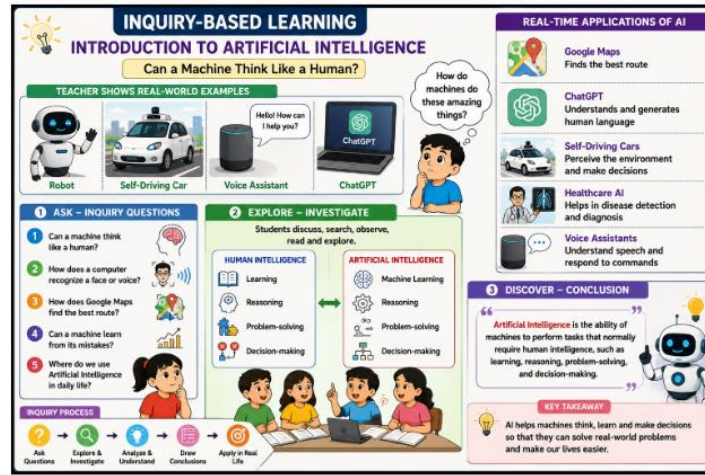
Artificial Intelligence (AI) is the field of computer science that enables machines and computer systems to perform tasks that normally require human intelligence.

In simple words:

AI makes machines think, learn, reason, solve problems, and make decisions like humans.

Examples of AI:

- **ChatGPT** – understands and generates human language.
- **Google Maps** – finds the best route.
- **Self-driving cars** – observe the environment and make driving decisions.
- **Voice assistants** – understand and respond to speech.
- **Medical AI** – helps detect diseases.
- **Chess AI** – selects the best move.





ASKING AI (The Idea) → AI MISTAKE (Too General) LEARNING TO BE SPECIFIC



AI Assistant:
My Creative Partner

AI Problems (Artificial Intelligence Problems)

Definition:

AI problems are the challenges or tasks that require a computer to imitate human intelligence, such as learning, reasoning, decision-making, understanding language, and recognizing images.

1. Bias and Fairness

Problem: AI may produce unfair or discriminatory results if trained on biased data.

Example:

A hiring AI trained mostly on resumes from men may reject qualified women candidates.

Solution:

- Use balanced datasets.
 - Regularly test AI for fairness.
-

2. Explainability and Transparency

Problem: Many AI models (especially deep learning) make decisions without explaining why.

Example:

A bank loan application is rejected by AI, but the customer does not know the reason.

Solution:

- Use Explainable AI (XAI).
 - Provide reasons for AI decisions.
-

3. Data Privacy and Security

Problem: AI requires large amounts of data, which may contain personal information.

Example:

A healthcare AI stores patient medical records. If hacked, private information may be exposed.

Solution:

- Encrypt data.
 - Follow privacy laws.
 - Limit access to sensitive data.
-

4. Ethical Concerns

Problem: AI may be used in ways that harm people or society.

Example:

Creating deepfake videos to spread false information.

Solution:

- Develop ethical AI guidelines.
 - Monitor AI usage responsibly.
-

5. Job Displacement

Problem: AI and automation can replace some human jobs.

Example:

Self-checkout systems reduce the need for cashiers.

Solution:

- Reskill and upskill workers.
 - Create new AI-related job opportunities.
-

6. Generalization and Robustness

Problem: AI may perform well on training data but fail in new or unexpected situations.

Example:

A self-driving car works well in sunny weather but struggles in heavy rain.

Solution:

- Train with diverse datasets.
 - Test AI under different conditions.
-

7. Scalability and Computational Resources

Problem: AI models require powerful computers, large memory, and high processing power.

Example:

Training a large language model may take weeks using expensive GPUs.

Solution:

- Use efficient algorithms.
 - Optimize AI models.
 - Use cloud computing.
-

8. Legal and Regulatory Issues

Problem: AI must follow laws related to privacy, safety, and copyright.

Example:

An AI system uses copyrighted images without permission.

Solution:

- Follow government regulations.
 - Respect copyright and privacy laws.
-

9. Human-AI Collaboration

Problem: AI should assist humans, not replace human judgment in critical situations.

Example:

A doctor uses AI to detect diseases but makes the final diagnosis.

Solution:

- Keep humans involved in important decisions.
- Use AI as a decision-support tool.

AGENTS AND ENVIRONMENTS

What is an Agent

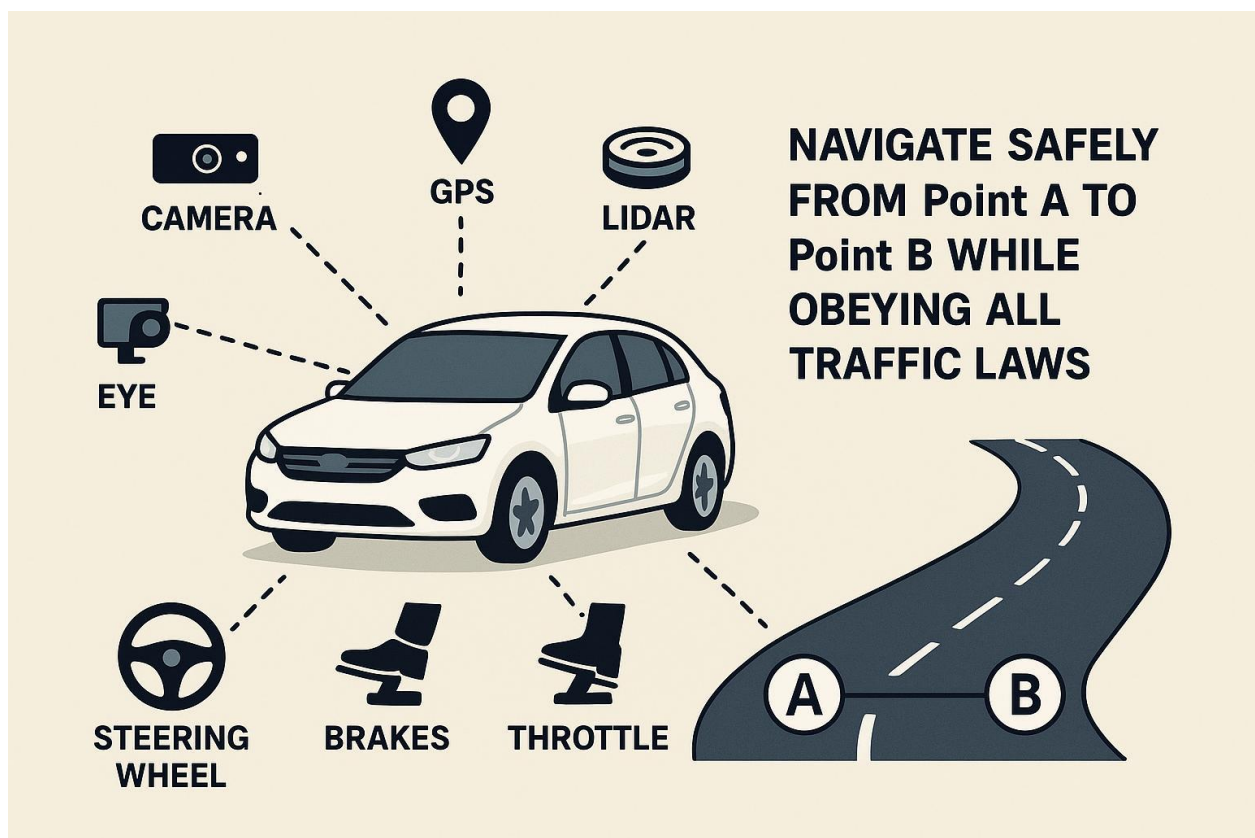
- In the context of Artificial Intelligence (AI), intelligent agents are systems that perceive their environment, make decisions based on that perception, and take actions to achieve specific goals.
- They are fundamental components of AI systems, which aim to mimic human intelligence by solving problems in dynamic environments.
- The agent operates autonomously, meaning it is not directly controlled by a human operator.

Characteristics of Agent

- **Perceiving its environment:** Using sensors (such as cameras, microphones, or software APIs for virtual agents) to gather information about its surroundings.
- **Taking actions:** Using actuators or output mechanisms (such as motors in robots, or software commands in virtual agents) to affect its environment.
- **Goal-Oriented Behavior:** An AI agent operates to achieve predefined goals or objectives based on the information it gathers from its environment.

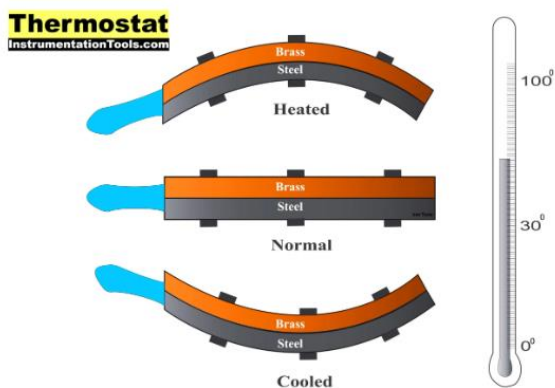
Example of Agent

- A self-driving car is an AI agent that perceives its environment (road conditions, traffic, pedestrians) using cameras and sensors.
- It processes this information to make decisions (steering, accelerating, braking) and acts accordingly to reach its destination safely.



Types of Agents

- There are different types of agents in AI, depending on how sophisticated they are:
 - Simple Reflex Agents
 - Model-Based Reflex Agents
 - Goal-Based Agents
 - Utility-Based Agents
- Simple Reflex Agents:
 - These agents make decisions based purely on the **current situation** and **predefined rules**.
 - They don't consider the history or future consequences of their actions.
 - Example: A thermostat that turns on the heater if the temperature drops below a certain point.



- **Model-Based Reflex Agents:**

- These agents maintain an internal state or model of the world, allowing them to handle partial observability and make decisions based on both current inputs and their history.
- Example: A robot vacuum cleaner that builds a map of a room as it cleans.

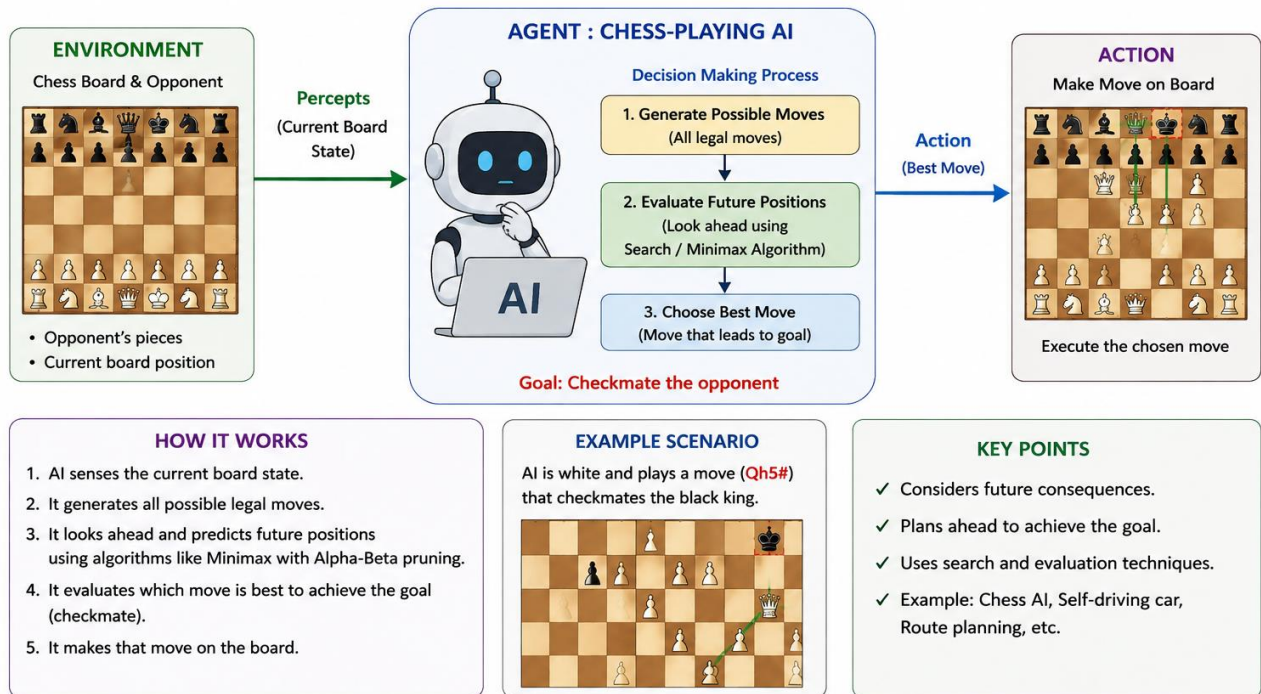


- **Goal-Based Agents:**

- These agents make decisions by considering future consequences and whether an action will help achieve a specific goal.
- Example: A chess-playing AI that evaluates possible future moves to checkmate the opponent.

Chess-Playing AI (Goal-Based Agent)

A chess-playing AI considers future consequences of moves and chooses the best move to achieve its goal – **CHECKMATE** the opponent.



- **Utility-Based Agents:**

- These agents go beyond simple goal achievement and try to maximize a utility function (a measure of success or satisfaction).
- They seek not only to reach a goal but to achieve the best possible outcome.
- Example: A self-driving car that not only reaches its destination but does so in the safest, quickest, and most comfortable way.

What is Environment

- The environment is the external world or context within which the agent operates.
- It provides input to the agent and reacts to the agent's actions.
- In AI, environments can vary greatly in complexity and predictability.

Types of Environment

- **Observable vs. Partially Observable:**
 - In an **observable environment**, the agent has full access to the relevant data (e.g., a chess game where the entire board is visible to the AI agent).
 - In a **partially observable environment**, the agent only has limited information (e.g., a robot navigating through a foggy landscape).

- **Deterministic vs. Stochastic:**

- In a **deterministic environment**, actions have predictable outcomes (e.g., moving a piece in chess always results in one position on the chess board).
- In a **stochastic environment**, actions may have unpredictable outcomes (e.g., driving in real-world traffic with unpredictable drivers).

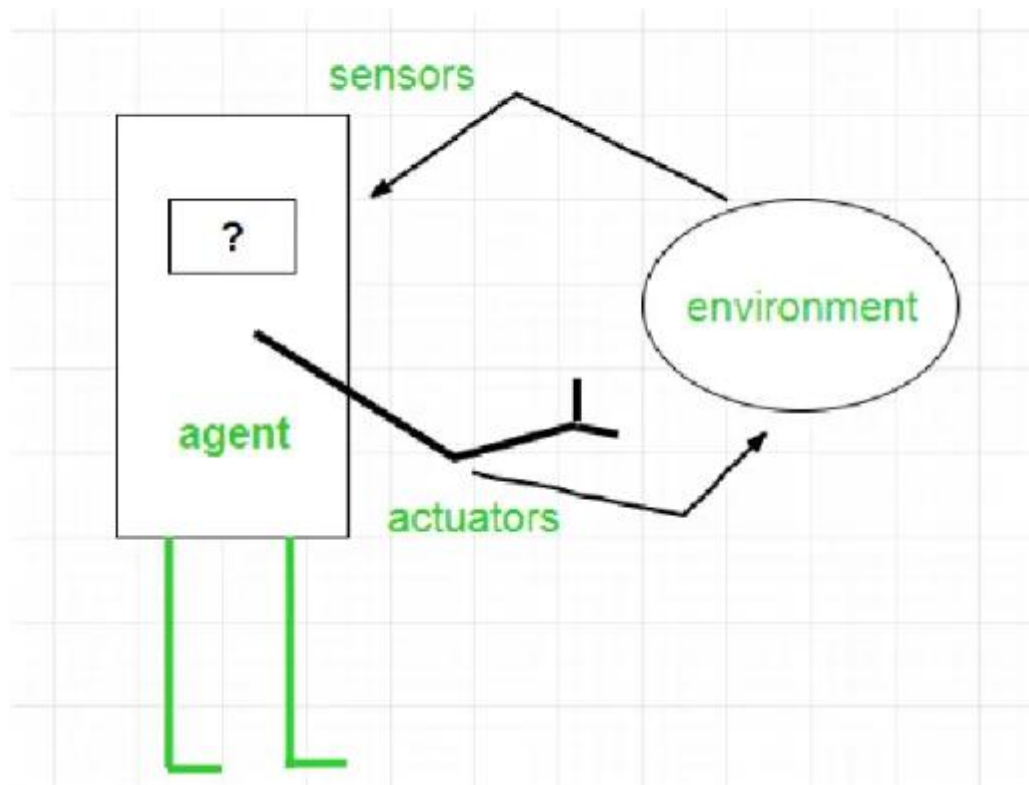
- **Static vs. Dynamic:**

- In a **static environment**, the world remains unchanged unless acted upon by the agent (e.g., a board game where nothing changes between moves).
- In a **dynamic environment**, the world can change independently of the agent's actions (e.g., a self-driving car dealing with changing traffic conditions).

- **Discrete vs. Continuous:**

- A **discrete environment** consists of distinct, countable states (e.g., a grid-based board game like tic-tac-toe).
- A **continuous environment** has a wide range of possible states and values (e.g., the position of a robot arm in space).

Agent & Environment



Nature of Environment and Agents

- **Nature of the Environment**

- Understanding the nature of the environment is crucial for designing an effective agent.
- Based on the environment's properties (deterministic vs. stochastic, static vs. dynamic, etc.), the agent must adapt its strategies.
 - In a **deterministic, fully observable** environment, an agent can plan its actions ahead with certainty (like playing chess).
 - In a **dynamic, stochastic, and partially observable** environment, an agent may need to rely on probabilistic models and adaptive strategies (like a drone navigating through unpredictable weather conditions).

- **Nature of the Agent**

- The design of an agent depends on the complexity of the problem it's solving and the nature of the environment it interacts with.
- The agent's **architecture** includes:
 - **Sensors** to perceive the environment.
 - **Effectors** or actuators to act on the environment.
 - **Decision-making algorithms** (such as search algorithms, learning algorithms, or rule-based systems) to choose appropriate actions.

Problem Solving Agent

- A **problem-solving agent** is a type of intelligent agent designed to find solutions to specific problems by exploring the **search space**, which contains all possible states or configurations.
- These agents aim to find a **sequence of actions** that leads to the desired goal state from a given initial state.

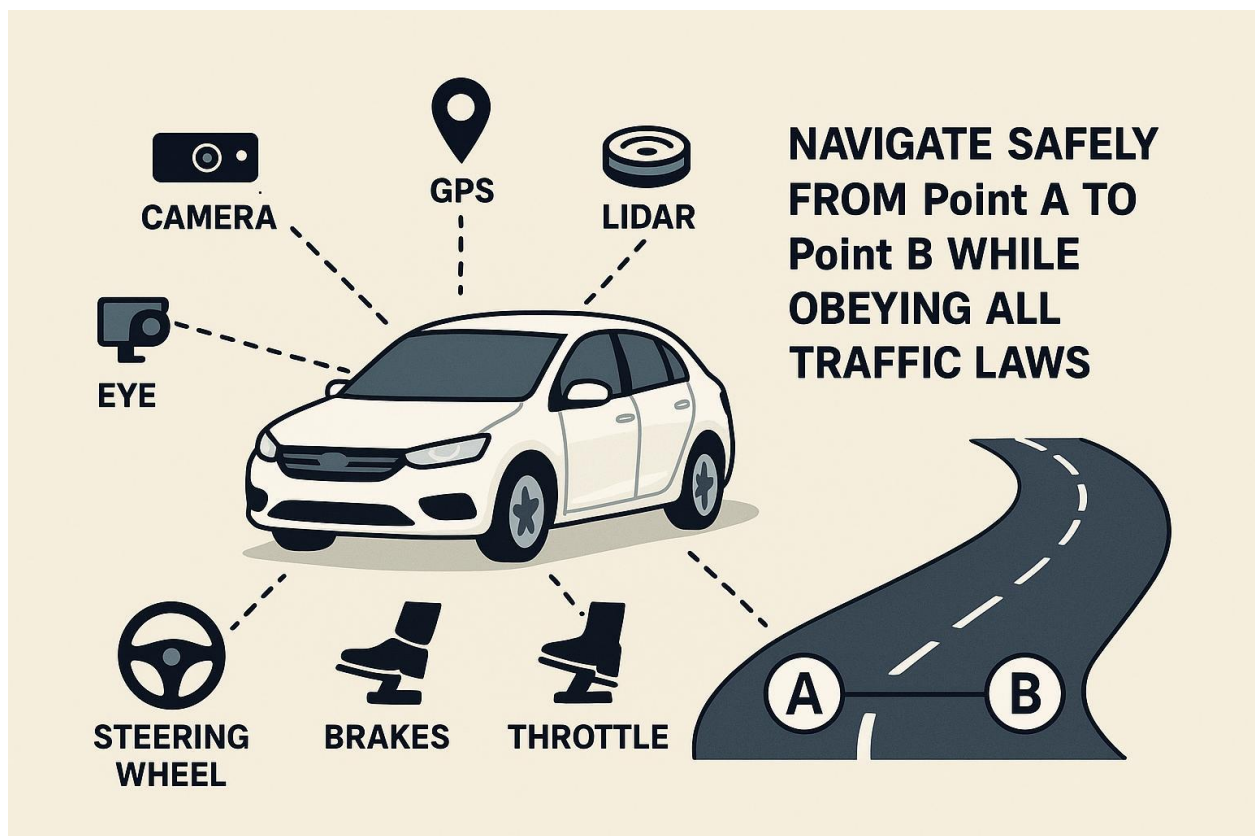
Steps in Problem Solving by an Agent

- **Formulate the Problem:**
 - The agent defines the initial state, the possible actions, and the goal state.
 - Example: In a maze-solving robot, the problem is to find the shortest path from the entrance to the exit.
- **Search for a Solution:**
 - The agent explores the search space, trying different sequences of actions to find the one that reaches the goal.
 - Example: The robot might use algorithms like depth-first search (DFS) or **A*** to explore all possible paths in the maze.
- **Execute the Solution:**
 - Once a solution is found, the agent executes the actions to achieve the goal.

Example of a Problem-Solving Agent in AI

- **Self-driving Car**

- **Environment:** Roads, traffic signals, other vehicles, pedestrians, weather conditions.
- **Agent:** The car itself, equipped with sensors (cameras, LIDAR) to perceive the environment and actuators to control its movement.
- **Problem-Solving:** The car's goal is to drive from point A to point B safely. It uses sensors to gather data, maps the environment, and uses algorithms (like A*) to plan the best route while avoiding obstacles.



CONCEPT OF RATIONALITY (ARTIFICIAL INTELLIGENCE)

Definition

The Concept of Rationality in Artificial Intelligence refers to an agent's ability to choose the action most likely to achieve its goal based on available information and knowledge.

Simple Definition

A rational agent does the right thing at the right time to achieve the best possible result.

Formula

Rational Action = Best Action + Available Knowledge + Current Percepts + Performance Measure

Working of Rationality

1. Agent receives information from the environment.
2. Analyzes the current situation.
3. Predicts possible outcomes.
4. Selects the action with the highest expected performance.
5. Executes the action.
6. Receives new percepts and repeats the process.

Examples

GPS Navigation, Warehouse Robot, Self-Driving Car.

Debate Activity

Topic: Should an AI agent always prioritize efficiency over safety?

Team A: Prioritize efficiency.

Team B: Prioritize safety.

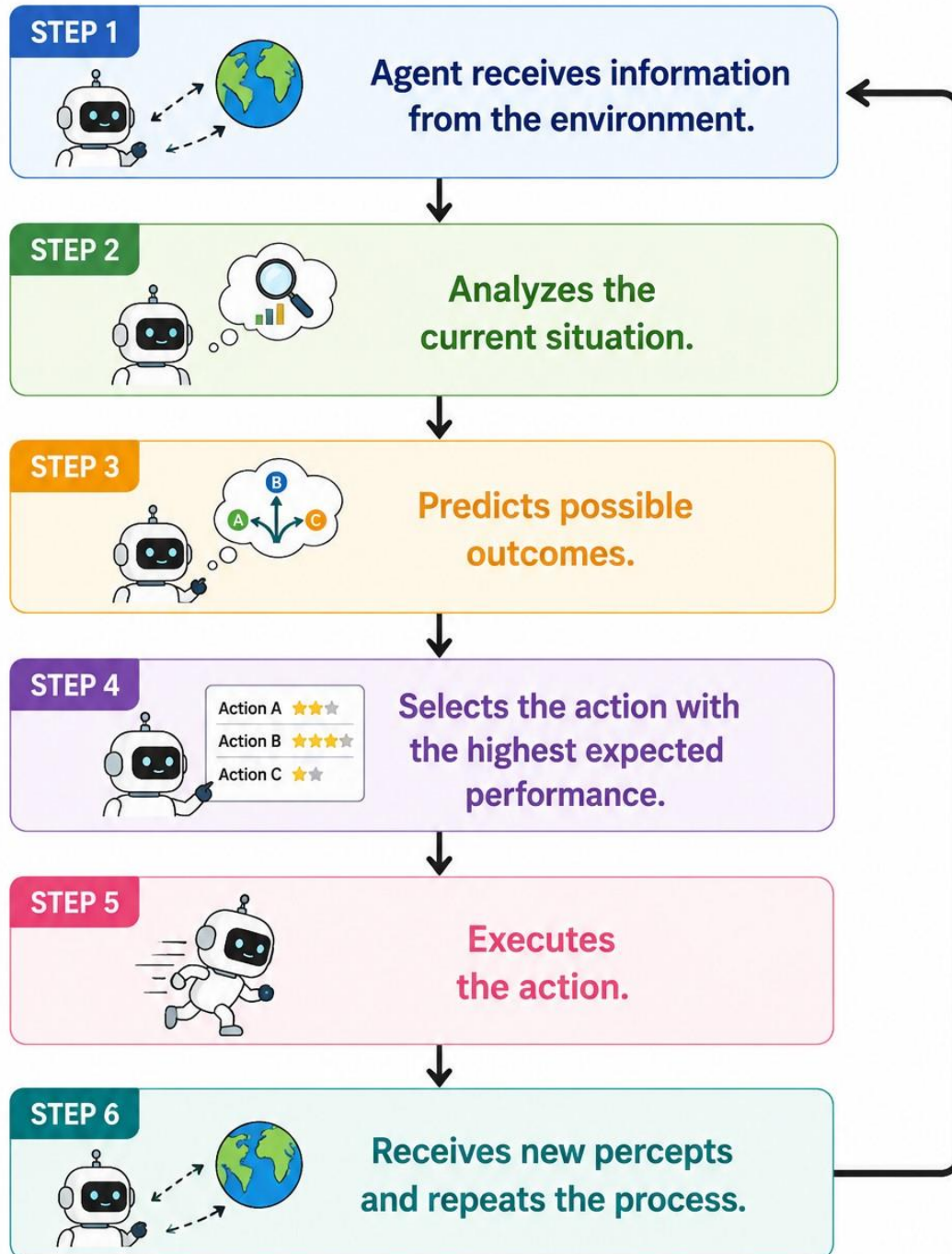
Discuss GPS Navigation, Warehouse Robots, and Self-driving Cars.

Characteristics of a Rational Agent

Characteristic	Description
Goal-Oriented	Works towards achieving a specific objective.
Intelligent Decision Making	Chooses the best action.
Uses Percepts	Uses sensor information.
Performance Driven	Maximizes performance.
Learns from Experience	Improves through learning.
Handles Uncertainty	Acts despite incomplete information.

Working of Rationality

A rational agent acts intelligently to achieve the best possible outcome.



Key Idea:

A rational agent continuously perceives, thinks, acts, and learns to achieve the best possible results.

The Concept of Rationality (AI)

Rationality means doing the right thing at the right time based on the available information.

In Artificial Intelligence, a rational agent is an agent that always chooses the action that maximizes its chances of achieving its goal.

A rational agent is an intelligent agent that selects the best possible action based on its **knowledge, percepts,** and **performance measure** to achieve its goal.

What is a Rational Agent?

A rational agent

- Observes its environment through sensors.
- Takes actions through actuators.
- Chooses the action that gives the best expected outcome.

What is a Rational Agent in AI?

- One of the core aspects of AI is its ability to make decisions and solve problems. This capability hinges on the concept of rationality.
 - A rational agent, often known as a rational being, is a person or entity that does the best actions possible given the circumstances and information at hand.
 - A rational agent can be any decision-making entity, such as a person, corporation, machine, or software.
 - A rational agent has a clear preference, models uncertainty, and acts in a way that maximizes its performance measure using all available actions.
-

Example 1: Robot Vacuum Cleaner

Situation:

- Room has dust.
- Battery is 80%.
- Dust is detected.

Rational Action:

The robot decides to clean the room because cleaning increases its performance.

If the battery becomes very low (5%), the rational action changes to going to the charging station. This is still rational because it helps achieve the overall goal.

Example 2: Self-Driving Car

The car detects:

- Traffic signal = Red
- Pedestrians crossing

A rational car stops at the signal because stopping best satisfies safety and traffic rules.

Example 3: Chess Program

The AI:

- Looks at all possible moves.

- Evaluates each move.
- Selects the move with the highest chance of winning.

That makes it a rational agent.

Types of Rationality

There are two primary types of rationality in AI: **bounded rationality** and **perfect rationality**.

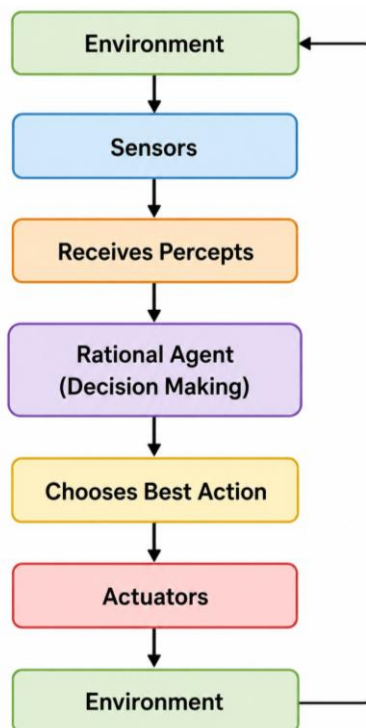
1. Bounded Rationality

Bounded rationality recognizes that decision-making capabilities are limited by the information available, cognitive limitations, and time constraints. AI systems operating under bounded rationality use heuristics and approximations to make decisions that are good enough, rather than optimal. This approach is practical in real-world applications where perfect information and infinite computational resources are unavailable.

2. Perfect Rationality

Perfect rationality assumes that an AI system has access to complete information, unlimited computational power, and infinite time to make decisions. While this is an idealized concept, it serves as a benchmark for evaluating the performance of AI systems. Perfectly rational AI would always make the best possible decision in any given situation.

How Does a Rational Agent Work?



- A rational agent is mainly used for goal-oriented agents. It evaluates its environment by examining what it is like.
- It then examines each available action in its armory to see how it will alter the surroundings and assist it achieve its goal.
- It tests all possible steps before selecting the best one, the one that would bring it closest to its goal.
- Rational agents are also used in self-driving cars, energy-saving air-conditioning units, automated lights, and other devices that require environmental data to choose the optimal course of action.

Criteria for Measuring Rationality

Criterion	Meaning	Example	Explanation
Performance Measure	Defines what counts as success for the agent.	Robot Vacuum: A clean room. Self-Driving Car: Reach the destination safely.	The agent chooses actions that maximize its performance. If the robot cleans more of the room, it has achieved a higher performance measure.
Percept Sequence	The complete history of information received through sensors.	Robot Vacuum: Detects dust, walls, and battery level. Self-Driving Car: Detects traffic lights, pedestrians, and nearby vehicles.	The agent uses all the information collected from its sensors to decide what action to take next.
Prior Knowledge	Information already stored in the agent before making a decision.	GPS Navigation: Digital road map. Warehouse Robot: Knows the warehouse layout and charging station location.	The agent combines its stored knowledge with current observations to make better decisions.
Available Actions	The set of actions the agent can perform.	Robot Vacuum: Move forward, turn left, turn right, clean, recharge. Self-Driving Car: Accelerate, brake, turn, stop.	The agent evaluates all possible actions and selects the one that gives the best expected outcome.

Achieving Rationality in AI

Implementing rationality in AI involves several techniques and approaches:

1. Decision Theory

Decision theory provides a framework for making rational choices by evaluating the potential outcomes of different actions. It combines probability theory and utility theory to calculate the expected utility of each action and select the one with the highest expected utility. This approach is widely used in AI for planning and problem-solving tasks.

2. Game Theory

[Game theory](#) studies strategic interactions between agents, where the outcome for each participant depends on the actions of others. In AI, game theory is used to model and analyze competitive and cooperative scenarios, enabling agents to make rational decisions in multi-agent environments.

3. Machine Learning

[Machine learning](#) algorithms enable AI systems to learn from data and improve their decision-making over time. By identifying patterns and relationships in data, machine learning models can make more accurate predictions and choose actions that maximize performance.

4. Logic and Reasoning

Logical reasoning involves using formal rules and knowledge representations to infer new information and make decisions. Techniques such as propositional logic, predicate logic, and [Bayesian networks](#) help AI systems reason about the world and make rational decisions based on the knowledge they possess.

5. Reinforcement Learning

[Reinforcement learning](#) is a type of machine learning where an agent learns to make decisions by interacting with an environment. The agent receives feedback in the form of rewards or penalties and uses this feedback to learn a policy that maximizes cumulative rewards. This approach is particularly effective for sequential decision-making problems.

Challenges in Achieving Rationality

While the goal of achieving rationality in AI is desirable, it comes with several challenges:

1. Uncertainty and Incomplete Information

In many real-world situations, AI systems must make decisions with incomplete or uncertain information. Handling uncertainty effectively requires sophisticated probabilistic reasoning and robust learning algorithms.

Example 1: Self-Driving Car ?

Situation:

A self-driving car is driving in heavy fog.

- The camera cannot clearly detect the road.
- Sensors may not accurately identify pedestrians or other vehicles.
- The AI does not have complete information about the surroundings.

How AI Handles It:

The AI combines data from multiple sensors such as:

- Camera
- Radar
- LiDAR
- GPS

It estimates the safest action, such as:

- Slowing down
- Maintaining a safe distance
- Turning on fog lights

Reason: The AI makes the best possible decision even though it cannot see everything clearly.

2. Computational Complexity

Optimal decision-making often involves searching through a vast space of possible actions and outcomes. This can be computationally expensive, especially for complex problems. Balancing the trade-off between computational resources and decision quality is a key challenge.

Example 2: Google Maps (GPS Navigation) □

Situation

Suppose you want to travel from **Home** to **College**.

There may be:

- Thousands of roads
- Hundreds of intersections
- Multiple possible routes

Checking every route would take too much time.

How AI Handles It

Google Maps:

- Uses efficient search algorithms.
- Quickly finds the shortest or fastest route.
- Updates the route if traffic changes.

Reason: Instead of examining every possible route, the AI searches efficiently to provide a good solution in seconds.

3. Ethical and Social Considerations

Rational decisions made by AI systems can have significant ethical and social implications. Ensuring that AI systems align with human values and do not cause harm requires careful consideration of ethical principles and societal impact.

Example 4: Medical Diagnosis System □

Situation

An AI system suggests that a patient has a disease.

If the prediction is incorrect, the patient may receive the wrong treatment.

How AI Should Handle It

The AI should:

- Provide accurate recommendations.
- Explain its reasoning when possible.
- Assist doctors rather than replace them.

Reason: AI should support healthcare professionals and reduce the risk of harmful decisions.

Applications of Rational AI

Rational AI has numerous applications across various domains:

1. Autonomous Vehicles

Autonomous vehicles use rational decision-making to navigate safely, avoid obstacles, and optimize routes. They must make real-time decisions based on sensor data and changing traffic conditions.

2. Healthcare

In healthcare, AI systems assist in diagnosis, treatment planning, and personalized medicine. Rational decision-making helps optimize patient outcomes by considering medical history, current symptoms, and treatment options.

3. Finance

AI is used in finance for algorithmic trading, risk management, and fraud detection. Rational AI systems analyze market data, predict trends, and make investment decisions to maximize returns.

4. Robotics

Robots in manufacturing, logistics, and service industries rely on rational decision-making to perform tasks efficiently and adapt to dynamic environments.

Rationality in Decision-Making

- **Research and brainstorm possible solutions:** The probability of addressing your problem increases when you broaden your pool of prospective solutions. To identify as many feasible answers as you can, you should research your problem thoroughly using both the internet and your knowledge. To come up with more ideas, brainstorming with others is another option.
- **Set standards of success and failure for your potential solutions:** Setting a threshold for measuring the success and failure of your ideas allows you to identify which ones will genuinely address your problem. However, your expectations of success should not be unrealistic. You'd be unable to discover a solution. However, if your standards are practical, quantitative, and targeted, you will be able to identify one.
- **Describe the potential outcomes of each solution:** The next step is to determine the repercussions of each of your solutions. Create a table of each alternative's strengths and shortcomings and compare them. You should also prioritize your solutions in a list, from greatest to worst possibility of solving the problem.
- **Choose the best solution and test it:** Choose the best solution and test it after evaluating all of your options. You can also begin to monitor your early results at this stage.
- **Implement the solution or try a new one:** If your suggested answer passed your test and solved your problem, it is the most logical choice you can make. You should use it to fix your current problem as well as any future ones that arise. If the answer did not fix your problem, try another possible solution that you came up with.

Four Factors That Determine Rationality

1. Performance Measure

Defines what counts as success.

Examples:

- Robot vacuum → Clean room
- Self-driving car → Reach destination safely
- Chess AI → Win the game

2. Percept Sequence

All the information the agent has received so far.

Example:

- Dust

- Wall
- Battery level

3. Prior Knowledge

Knowledge already stored in the agent.

Example:

- Charging station location
- Previously cleaned areas

4. Available Actions

Examples:

- Move forward
- Turn left
- Turn right
- Stop
- Pick up object
- Recharge

Characteristics of a Rational Agent

- Makes intelligent decisions.
- Uses available information.
- Tries to maximize success.
- Learns from experience (if possible).
- Does not always guarantee success but chooses the best expected action.

Rationality vs Omniscience

Rational Agent	Omniscient Agent
Uses available information	Knows everything
Chooses the best expected action	Always knows the correct action
May make mistakes if information is incomplete	Never makes mistakes
Practical in AI	Impossible in real life

Example:

A self-driving car cannot know that a tire will burst in the next minute. It drives normally based on available information, so it is still rational even though it cannot predict the future.

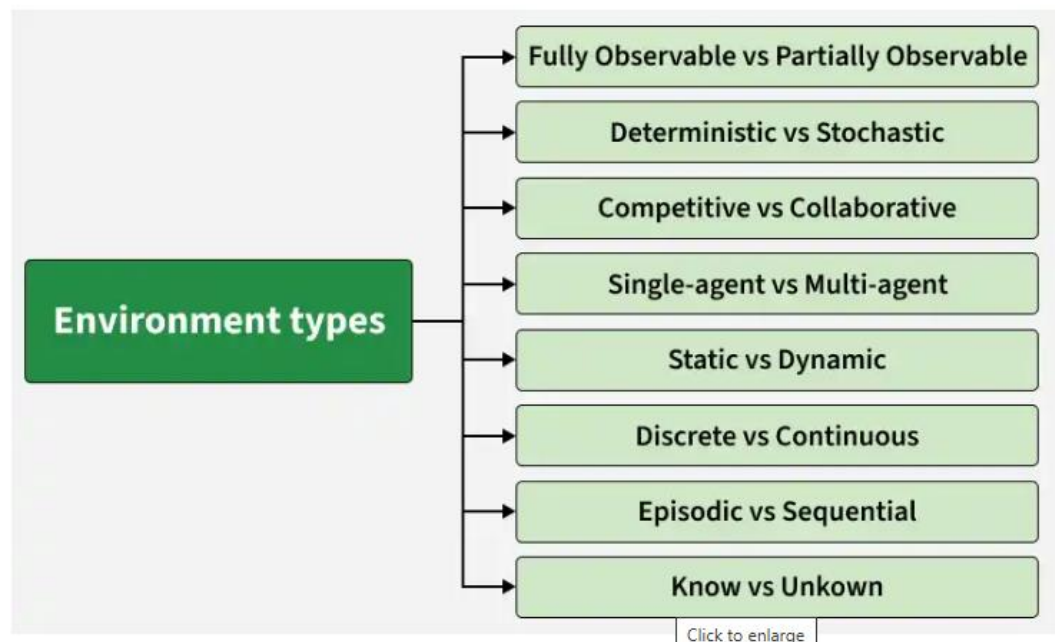
Nature of Environment in Artificial Intelligence

An environment in artificial intelligence refers to the external world in which an agent operates and interacts. The behavior of an environment depends on how much information the agent can access and how the state changes over time.

- The agent perceives information from the environment using sensors and performs actions through actuators.
- Environments are categorized based on interaction style, uncertainty, and structure of decision-making.

Component	Examples	Function
Sensors (Percepts)	Infrared (IR) Sensor	Detects obstacles such as walls and furniture.
	Ultrasonic Sensor	Measures the distance to nearby objects.
	Bumper Sensor	Detects collisions with objects.
	Cliff Sensor	Detects stairs or edges to prevent falling.
	Dirt Sensor	Identifies dusty or dirty areas that need extra cleaning.
	Wheel Encoder	Measures wheel rotation to estimate movement and distance.
	Camera / LiDAR (in advanced models)	Creates a map of the room and identifies obstacles.
	Battery Sensor	Monitors battery level for charging decisions.

Actuators (Actions)	Function
Drive Motors	Move the vacuum forward, backward, left, and right.
Wheel Motors	Rotate the wheels for navigation.
Suction Motor	Generates suction to collect dust and debris.
Rotating Brush Motor	Spins the brushes to sweep dirt into the suction inlet.
Side Brush Motor	Cleans corners and edges.
Speaker / Buzzer	Produces voice messages or alert sounds.
LED Display / Indicator	Displays battery status, errors, and cleaning mode.
Docking Mechanism	Connects to the charging station automatically.



Types of Environments

1. Fully Observable vs Partially Observable

In a fully observable environment, the agent can access the complete state at any given time, while in a partially observable environment, only partial information is available. Fully observable systems do not require maintaining historical data, whereas partially observable systems depend on memory of past states.

Examples:

- Chess - The board is fully observable, and so are the opponent's moves.
- Driving - The environment is partially observable because what's around the corner is not known

2. Deterministic vs Stochastic

When a uniqueness in the agent's current state completely determines the next state of the agent, the environment is said to be deterministic. The stochastic environment is random in nature which is not unique and cannot be completely determined by the agent.

Examples:

- Chess - There would be only a few possible moves for a chess piece at the current state and these moves can be determined.
- Self-Driving Cars- The actions of a self-driving car are not unique, it varies time to time.

3. Competitive vs Collaborative

In a competitive environment, agents compete against each other to optimize their individual outcomes. An agent is said to be in a collaborative environment when multiple agents cooperate to produce the desired output.

Examples:

- Chess - Each agent competes to defeat the opponent and win the match.
- Self-Driving Cars - Vehicles coordinate with each other to ensure safe travel and avoid collisions.

4. Single-agent vs Multi-agent

An environment consisting of only one agent is said to be a single-agent environment. A multi-agent environment involves more than one agent interacting within the same environment.

Examples:

- Maze solving – One agent navigates to find the exit.
- Football – Multiple players from both teams interact to play the game.

5. Dynamic vs Static

A dynamic environment keeps changing while the agent is performing actions or interacting with it. A static environment remains unchanged while the agent is acting on it.

Examples:

- Roller coaster ride – The state keeps changing continuously during motion.
- Empty house – Surroundings do not change when an agent enters.

6. Discrete vs Continuous

If an environment consists of a finite number of actions that can be deliberated in the environment to obtain the output, it is said to be a discrete environment. A continuous environment has uncountable or infinitely varying actions and states.

Examples:

- Chess – It has a limited number of possible moves in any given position.
- Self-driving cars – Actions like steering and acceleration vary continuously and cannot be counted.

7. Episodic vs Sequential

An episodic environment consists of independent tasks where each action is separate and does not depend on previous actions. A sequential environment involves dependent actions where current decisions affect future outcomes.

Examples:

- Pick and place robot – Each item is checked independently on a conveyor belt.
- Checkers – Each move influences all subsequent moves in the game.

8. Known vs Unknown

A known environment is one where the outcomes of all possible actions are already known to the agent. An unknown environment is one where the agent does not know the outcomes and must learn or explore to understand them.

Examples:

- Route planning in a GPS navigation system – Outcomes of paths are already known.
- Exploring an unfamiliar maze – The agent must discover paths and outcomes while navigating.

Nature of Environment in Artificial Intelligence

A Digital Story

1 What is the Environment in AI?

Everything around an agent with which it interacts is its environment!

2 The environment has different natures. Let's explore them with examples!

Observability
(Full / Partial)

Deterministic
or Stochastic

Static or
Dynamic

Episodic or
Sequential

Single-agent or
Multi-agent

3 1) Observability

Fully Observable

I can see the whole board! Nothing is hidden.

Example: Chess

Partially Observable

I cannot see the atmosphere completely. I have to predict.

Example: Weather Forecasting

4 2) Deterministic or Stochastic

Deterministic

Same action always gives same result.

Example: Calculator

Stochastic

Outcome is uncertain; it may change.

Example: Stock Market

5 3) Static or Dynamic

Static

Environment does not change while agent decides.

Example: Crossword Puzzle

Dynamic

Environment keeps changing with time.

Example: Traffic on Road

6 4) Episodic or Sequential

Episodic

Each experience is independent.

Example: Spam Email Detection

Sequential

Current action affects future actions.

Example: Robot Navigation

7 5) Discrete or Continuous

Discrete

Limited number of separate values.

Example: Traffic Light (Red, Yellow, Green)

Continuous

Infinite number of values in a range.

Example: Temperature in Celsius

8 6) Single-agent or Multi-agent

Single-agent

Only one agent operates.

Example: Automatic Vacuum Cleaner

Multi-agent

Many agents interact and affect each other.

Example: Self-driving Cars on the same road

9 Real-World Examples Bringing It All Together

Weather Forecasting

- Partially Observable
- Stochastic
- Dynamic
- Sequential
- Continuous
- Single-agent

HOLLY AI Trading Bot

- Partially Observable
- Stochastic
- Dynamic
- Sequential
- Continuous
- Multi-agent

Moral of the Story

Understanding the nature of environment helps AI agents make better decisions and achieve their goals in the real world!

Remember

Environment defines the challenges. Good AI understands the environment and acts intelligently to solve problems!

Remember

Environment defines the challenges. Good AI understands the environment and acts intelligently to solve problems!

Structure of Agents (Artificial Intelligence)

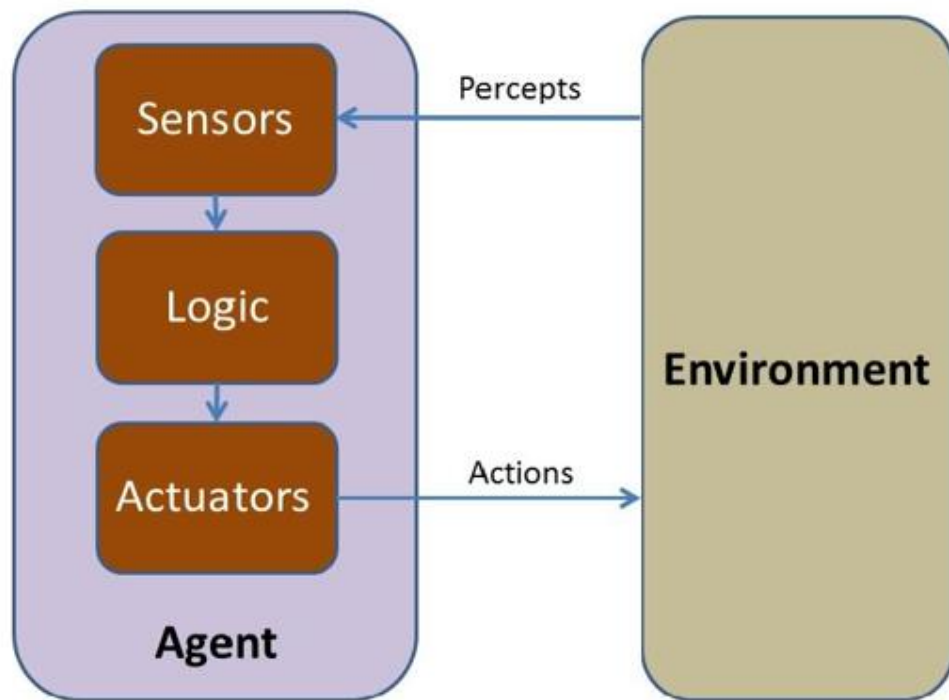
Definition

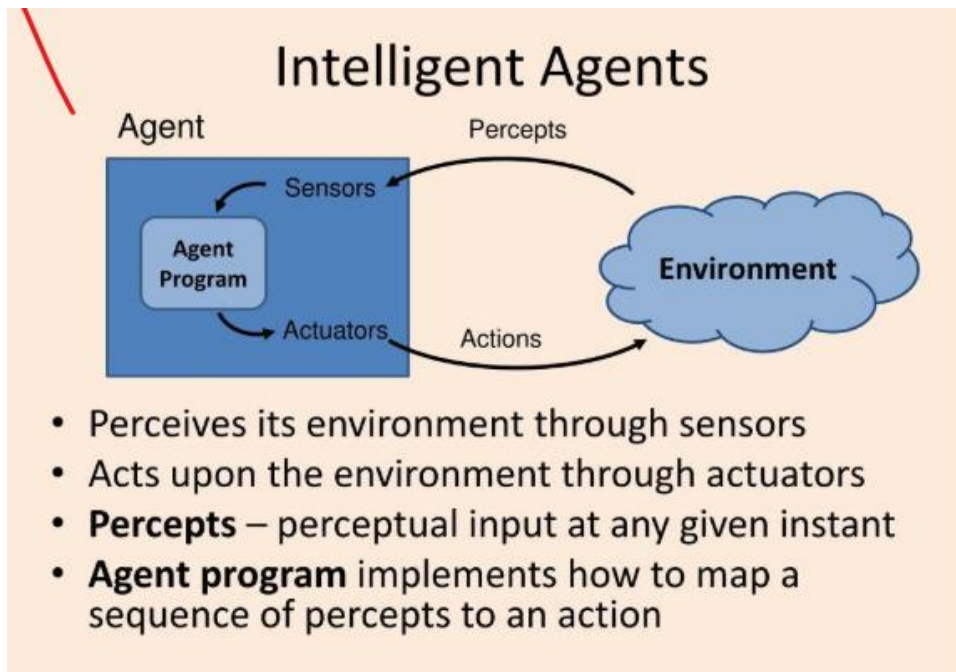
The **structure of an agent** describes how an intelligent agent is organized to perceive its environment, process information, and perform actions to achieve its goals.

An intelligent agent consists of two main parts:

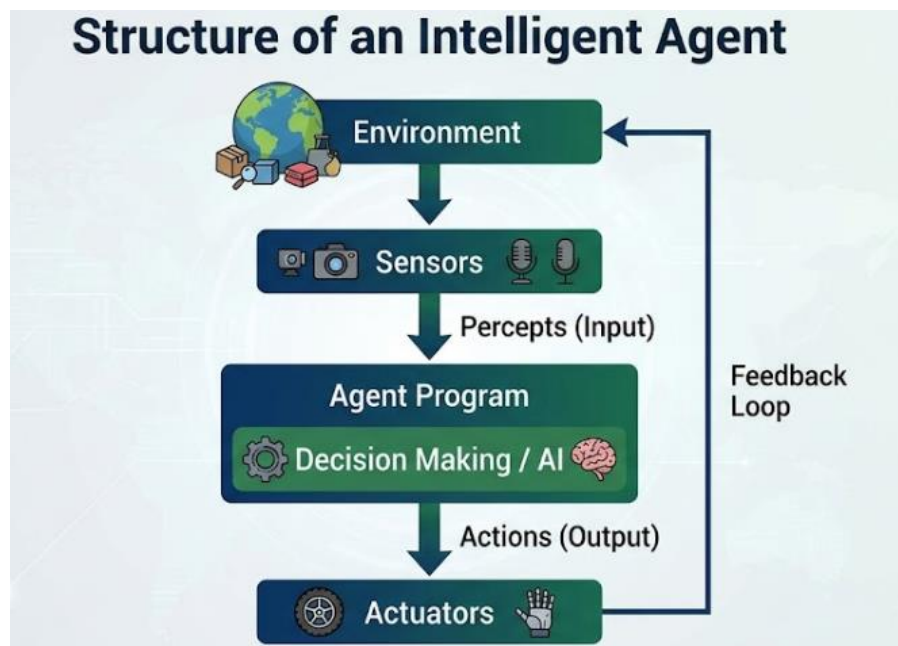
Agent = Architecture + Agent Program

- **Architecture:** The hardware or software platform on which the agent runs.
- **Agent Program:** The algorithm that maps percepts (inputs) to actions (outputs).
-





Components of the Structure



1. Environment

The surroundings in which the agent operates.

Examples

- Factory
 - Farm
 - Hospital
 - Home
 - Road
-

2. Sensors

Sensors collect information from the environment.

Examples

- Camera
- Microphone
- GPS
- Temperature sensor
- Moisture sensor
- LIDAR

Function

- Detect changes in the environment.
 - Send percepts to the agent program.
-

3. Agent Program

The agent program is the **brain** of the intelligent agent.

Functions

- Receives percepts.
 - Processes information.
 - Makes decisions.
 - Chooses the best action.
-

4. Architecture

The architecture provides the platform on which the agent program runs.

Examples

- Computer
 - Robot controller
 - Embedded system
 - Smartphone
 - Cloud server
-

5. Actuators

Actuators execute the decisions made by the agent program.

Examples

- Robot arm
 - Wheels
 - Motors
 - Sprayer
 - Speaker
-

Working of an Intelligent Agent

1. The **environment** generates information.
2. **Sensors** observe the environment.
3. The **agent program** processes the percepts.
4. The **architecture** executes the program.
5. **Actuators** perform the selected action.
6. The environment changes, and the cycle repeats

Real-Time Application 1: Industrial Automation



Component	Example
Environment	Factory assembly line
Sensors	Camera, proximity sensor
Agent Program	Quality inspection and control software
Architecture	Industrial robot controller
Actuators	Robotic arm, conveyor motor
Goal	Assemble products accurately and efficiently

Working

- Camera detects the product.
- The controller checks its position.
- The robot arm picks and places the component.
- The assembly process continues automatically.

Real-Time Application 2: Smart Agriculture Drone



Component	Example
Environment	Agricultural field
Sensors	Camera, GPS, moisture sensor
Agent Program	Crop health analysis software
Architecture	Drone onboard computer
Actuators	Propellers and spray nozzle
Goal	Monitor crops and spray only affected areas

Working

- Sensors capture crop images and moisture levels.
- AI identifies unhealthy crops.
- The drone sprays fertilizer or pesticide only where required.
- Healthy crops are left untouched, reducing waste.

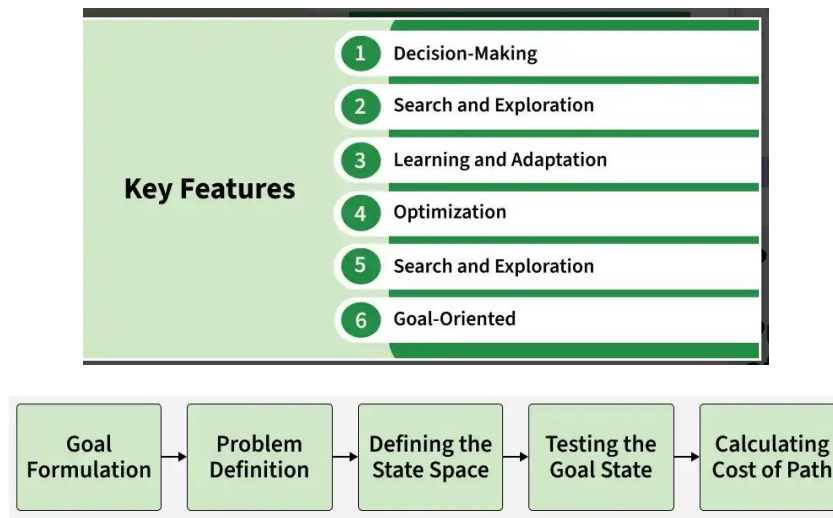


Problem-Solving Agents

Definition: A problem-solving agent is a goal-based intelligent agent that formulates a problem and searches for a sequence of actions that leads from the initial state to the desired goal state.

Problem-solving in Artificial Intelligence (AI) is the process of identifying a goal, analyzing available options, and selecting the most effective sequence of actions to achieve the desired outcome. It enables intelligent systems to reason, search for solutions, and make decisions in complex situations.

- Enables AI agents to make goal-oriented decisions in different environments.
- Forms the foundation of applications such as robotics, planning, navigation, and game playing.



How a Problem-Solving Agent Works

- **Goal Formulation:** The AI identifies the objective it needs to achieve. A clearly defined goal helps the agent focus on the desired outcome and guides the problem-solving process.
- **Problem Definition:** The problem is clearly specified by identifying the initial state, goal state, constraints, and available actions. This establishes the framework for finding a solution.
- **Defining the State Space:** The AI represents all possible states and transitions that can occur while solving the problem. This helps the agent explore different paths toward the goal.
- **Testing the Goal State:** The system checks whether the current state satisfies the desired goal conditions. If the goal is reached, the search process can terminate.

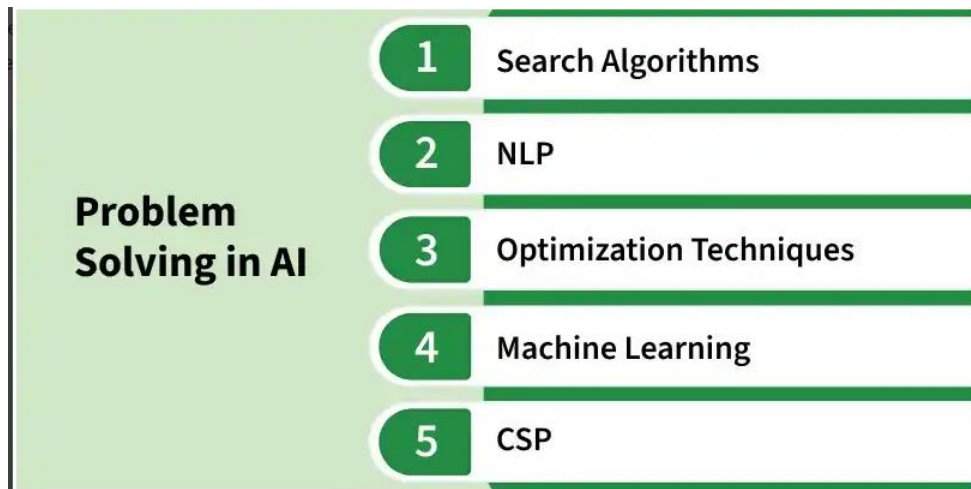
- **Calculating Cost of Path:** The AI evaluates the cost associated with different solution paths, such as time, distance, or resources, and selects the most efficient path.

Characteristics of Problem-Solving Agents

Feature	Description
Goal-oriented	Works toward a specific goal
State-based	Represents the problem using states
Search-based	Searches for possible solutions
Action-oriented	Performs actions to change states
Solution sequence	Produces a sequence of actions
Cost consideration	Can select a low-cost solution

Overall Process

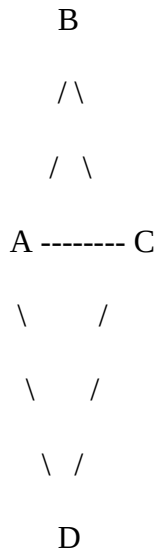
Goal Formulation → Problem Formulation → Search → Solution → Execution



1. Search Algorithms

Search algorithms are techniques used in Artificial Intelligence to explore a set of possible states or solutions and find a path from an **initial state** to a **goal state**.

For example, suppose a robot has to travel from **Room A to Room D**. There may be several possible routes:



2. Constraint Satisfaction Problems (CSP)

In [CSPs](#) the goal is to find solutions that satisfy a set of constraints. Techniques like backtracking, constraint propagation and local search are used to explore the solution space efficiently while ensuring that all constraints are met.

3. Optimization Techniques

AI often addresses optimization problems where the goal is to find the best possible solution among many options. Techniques include:

1. [Linear programming](#) for solving problems involving linear relationships.
2. [Dynamic programming](#) for breaking down complex problems into smaller sub problems.
3. [Evolutionary algorithms](#) like genetic algorithms that simulate natural selection to find optimal solutions.

4. Machine Learning

[Machine learning](#) allows AI systems to learn from data and improve over time. The three major learning types are:

1. [Supervised Learning](#): Learning from labelled data to make predictions or classifications.
2. [Unsupervised Learning](#): Discovering patterns in data without labeled outputs.
3. [Reinforcement Learning](#): Learning through trial and error, receiving feedback in the form of rewards or penalties.

5. Natural Language Processing (NLP)

[Natural Language Processing \(NLP\)](#) helps AI systems understand and generate human language. Techniques like tokenization, Part-of-speech (PoS) and named entity recognition are important for chatbots, language translation and text summarization.

Example: Vacuum Cleaner Agent

Suppose a vacuum cleaner is in Room A and the dirt is in Room B.

Component	Description
Initial state	Vacuum is in Room A.
Goal	Clean Room B.
Actions	Move right, suck dirt.
Search	Find a sequence of actions.
Solution	Move → Suck.
Goal achieved	Room B is clean.

Applications

- **Robotics and Navigation:** Enables robots to navigate environments and perform tasks autonomously.
- **Game Playing:** Helps AI analyze strategies and make optimal decisions in games.
- **Medical Diagnosis:** Assists in identifying diseases by analyzing medical data and symptoms.
- **Route Planning and Logistics:** Finds efficient routes to reduce travel time and operational costs.
- **Recommendation Systems:** Suggests relevant products, services, or content based on user preferences.

Advantages

- **Efficiency:** Solves problems quickly by processing large amounts of data.
- **Accuracy:** Produces reliable results when trained with quality data.
- **Automation:** Reduces human effort by handling repetitive tasks automatically.
- **Adaptability:** Learns from new data and improves performance over time.
- **Cost-Effectiveness:** Increases productivity and reduces operational costs.

Challenges

- **Complexity:** Some problems require significant computational resources.
- **Data Quality:** Poor or biased data can affect decision accuracy.
- **Interpretability:** Advanced AI models can be difficult to understand and explain.
- **Ethics and Bias:** AI systems may inherit biases present in training data.

PROBLEM FORMULATION

In artificial intelligence (AI) and machine learning, an agent is an entity that perceives its environment, processes information and acts upon that environment to achieve specific goals. The process by which an agent formulates a problem is critical, as it lays the foundation for the agent's decision-making and problem-solving capabilities.

Understanding Problem Formulation

Problem formulation is the process by which an [agent](#) defines the task it needs to solve. This involves specifying the initial state, goal state, actions, constraints, and the criteria for evaluating solutions. Effective problem formulation is crucial for the success of the agent in finding optimal or satisfactory solutions.

Steps in Problem Formulation

- 1. Define the Initial State:** The initial state is the starting point of the agent. It includes all the relevant information about the environment that the agent can perceive and use to begin the problem-solving process.
 - **Example:** In a navigation problem, the initial state could be the agent's starting location on a map.
- 2. Specify the Goal State:** The goal state defines the desired outcome that the agent aims to achieve. It represents the condition or [set](#) of conditions that signify the completion of the task.

- **Example:** For the navigation problem, the goal state is the destination location.

3. **Determine the Actions:** Actions are the set of operations or moves that the agent can perform to transition from one state to another. Each action should be well-defined and feasible within the given environment.

- **Example:** In a robot navigation scenario, actions could include moving forward, turning left, or turning right.

4. **Establish the Transition Model:** The transition model describes how the environment changes in response to the agent's actions. It defines the rules that govern state transitions.

- **Example:** In a game, the transition model would include the rules that specify how the game state changes based on the player's moves.

5. **Set Constraints and Conditions:** Constraints are the limitations or restrictions within which the agent must operate. These can include physical limitations, resource constraints, and safety requirements.

- **Example:** For a delivery drone, constraints might include battery life, weight capacity, and no-fly zones.

6. **Define the Cost Function (if applicable):** The cost function evaluates the cost associated with different

actions or paths. It helps the agent to optimize its strategy by minimizing or maximizing this cost.

- **Example:** In route planning, the cost function could represent the distance traveled, time taken, or energy consumed.

7. Criteria for Success: The criteria for success determine how the agent evaluates its progress and final solution. This includes metrics for measuring the effectiveness and efficiency of the solution.

- **Example:** For a puzzle-solving agent, success criteria could be the completion of the puzzle within the shortest time or the fewest moves.

Example of Problem Formulation in AI

Example: Finding a Route from Hyderabad to Bengaluru

Suppose an **AI agent** needs to find the best route from **Hyderabad to Bengaluru**.

1. Initial State

Where does the agent start?

Hyderabad

2. Actions

What can the agent do?

The agent can travel from one connected city to another.

For example:

Hyderabad→Kurnool Hyderabad→Anantapur

3. Transition Model

What happens after an action?

If the agent travels from Hyderabad to Kurnool:

Result(Hyderabad,Travel)=Kurnool

4. Goal Test

How does the agent know it has reached the goal?

Current City = Bengaluru

If the current city is Bengaluru → **Goal achieved.**

5. Path Cost

What is the cost of the solution?

The cost could be:

- Distance
- Travel time
- Fuel consumption
- Money

For example:

Hyderabad→Kurnool→Anantapur→Bengaluru

Total cost:

300+350+210=860 km

The AI agent tries to find a route with **minimum cost.**

Complete Problem Formulation

Component	Example
Initial State	Hyderabad
Actions	Travel to connected cities
Transition Model	Moving from one city to another
Goal Test	Current city = Bengaluru
Path Cost	Total distance/time/fuel

